

The empirical recurrence for OEIS A237170: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A237170 records “Empirical recurrence of order 75”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 202$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 75, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 75 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237170 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1696, 25384, 370670, 5876404, 93633848, 1522516754, 24903561376, 410933835890, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:55:42, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 202$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 202$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 404$ exact terms of the model returns order 75 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	42	c_{20}	−4821647012167932615	c_{39}	−2091156696388820069165086299	c_{58}
c_2	−200	c_{21}	−35230672810770863020	c_{40}	986150282278397970470985215	c_{59}
c_3	−13865	c_{22}	119815925611797339580	c_{41}	5550957861367845992587119606	c_{60}
c_4	178231	c_{23}	399302765019585986967	c_{42}	−5465127455770288417999745966	c_{61}
c_5	1582899	c_{24}	−2034763208847077077320	c_{43}	−10711476514584112231842228080	c_{62}
c_6	−36847314	c_{25}	−2952981664132399365163	c_{44}	17224727992861766578089894636	c_{63}
c_7	−30022430	c_{26}	25557714831009194351010	c_{45}	13454189450330305517181041764	c_{64}
c_8	3930027822	c_{27}	7597196816489732891319	c_{46}	−37581411171840085624805802584	c_{65}
c_9	−10796333296	c_{28}	−244421424272298925917771	c_{47}	−5779357774683243988393019520	c_{66}
c_{10}	−254160338775	c_{29}	133423378126754223121142	c_{48}	59049871099596272036424832608	c_{67}
c_{11}	1351226532795	c_{30}	1792797617824200067535664	c_{49}	−16596664039327904452543332384	c_{68}
c_{12}	10423480254120	c_{31}	−2285976172502440423434913	c_{50}	−66087286624193446947493541376	c_{69}
c_{13}	−85854647426135	c_{32}	−9977303805235283369948651	c_{51}	44218230907086633175529522368	c_{70}
c_{14}	−256266292535981	c_{33}	20469029382758672005780330	c_{52}	49355354191342837407697083776	c_{71}
c_{15}	3562189924627891	c_{34}	40437055890040082135829545	c_{53}	−57773689016050705899461247488	c_{72}
c_{16}	2181601254733383	c_{35}	−127693695090293973040308166	c_{54}	−19076059128455746505837351424	c_{73}
c_{17}	−104444678293908555	c_{36}	−104040805612161854388749529	c_{55}	47908113689601352879978061824	c_{74}
c_{18}	96643291574413178	c_{37}	593392715973637567424597362	c_{56}	−3860759762679173282170552320	c_{75}
c_{19}	2236393294416712573	c_{38}	49460050876741302982881651	c_{57}	−25656404994628140323801157632	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 75, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $75 + 4$ terms removed returns order 75 as well, so $\deg q_{\min} = 75 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $75 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 75 that this sequence satisfies — and that family turns out to have exactly one member.

References

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